

Stepping into the Operating Theater: ARAV - Augmented Reality Aided Vertebroplasty

Christoph Bichlmeier¹* Ben Ockert²† Sandro Michael Heining² Ahmad Ahmadi¹ Nassir Navab¹

¹Computer Aided Medical Procedures & Augmented Reality (CAMP), Technische Universität München, Germany

²Trauma Surgery Department, Klinikum Innenstadt, LMU München, Germany

ABSTRACT

Augmented Reality (AR) for preoperative diagnostics and planning, intra operative navigation and postoperative follow-up examination has been a topic of intensive research over the last two decades. However, clinical studies showing AR technology integrated into the real clinical environment and workflow are still rare. The incorporation of an AR system as a standard tool into the real clinical workflow has not been presented so far. This paper reports on the strategies and intermediate results of the *ARAV - Augmented Reality Aided Vertebroplasty* project that has been initiated to make an AR system based on a stereo video see-through head mounted display that is permanently available in the operating room (OR).

Index Terms: H.5.1 [Information Interfaces and Presentation]: Artificial, augmented, and virtual realities—; J.3 [Life and Medical Sciences]

1 INTRODUCTION

So far, only a small number of clinical studies report on the integration and evaluation of Augmented Reality (AR) systems inside an operating room (OR) [5, 3, 4]. Most of these approaches are add-on solutions for products that have already been certified, integrated and accepted for use in the OR. However, innovation and creativity can be restricted due to the implicated limitations of an existing product. This research follows the strategy of investigating a system design together with surgeons that takes full advantage of AR in order to determine its potentials in the OR.

2 TRANSFER PREPARATION OF THE AR SYSTEM

The following sections report on the preparative steps, the current state and future tasks for transferring a AR system [2] from lab space to the OR. The present AR system follows the concept of video see-through hardware setups that have been determined as the most promising approach regarding time synchronization, image composition and pose registration of the real and virtual world [8]. The video see-through head mounted display (HMD) has been evaluated in a first trail of experiments including an animal study for needle biopsy in an operating room equipped with a MRI scanner [9]. However, no patient study or further concepts for the integration of this system into the OR have been reported so far. The system is capable of tracking rigid targets in our setup with an accuracy of RMS 0.3mm. However, provision of a similar registration quality for deformable anatomy is still a topic of active research. One important issue for an effective environment to test medical AR systems is a realistic simulation of human anatomy. Cadaver and animal studies come close to the real conditions in the OR, however,

both are hard and expensive to be prepared. For this reason, we created the *Visible Korean Human Phantom* (VKHP) (Fig. 1(c)) from real imaging data taken from the *Visible Korean Human* (VKH) [7] full body CT imaging data set. The data of the VKHP was prepared with the software 3-Matic and Mimics of Materialise Group and printed with selective laser sintering by the company FIT Fruth Innovative Technologien GmbH. The surgeon is provided with parameterized AR views in particular stages of the procedure where the outcome of the intervention can be actually improved due to the extended AR vision. For this reason we aim at incorporating knowledge about surgical activity into workflow-driven AR solutions [6]. Based upon discussions with our medical partners, we have identified several potential benefits of workflow-driven AR in vertebroplasty, which is a minimal invasive spine surgery with the objective to insert bone cement into weak and brittle vertebrae through a *trokar* for stabilization. First, the trokar has to be positioned dorsal above the target vertebra and then inserted by hammering. Second, bone cement is carefully injected through the trokar. Too little bone cement would leave the vertebra in an unstable condition that potentially requires an additional surgery for reoperation. Too much bone cement would affect, in the worst case strangulate, sensitive nerves and vessels around and within the spinal canal. Physicians control and discuss each step with frequently updated x-ray scans that are presented on a monitor. The most critical phases, the penetration of vertebrae with the trokar and the injection of bone cement, can be supported with intelligently adapted AR visualization. Within the scope of a recent master thesis [1], various methods for activity discovery and classification in vertebroplasty have been investigated. For instance, accelerometers were used to capture surgical motion since we hypothesize that particular work steps correlate well with specific surgical movements (e.g. hammering, bone cement stirring, silence, waiting). We developed the new concept of the *shadow surgeon*, which is a member of the clinical staff, for instance an assisting surgeon or a medical student, watching the intervention through the HMD without actively contributing to the operation and influencing its outcome. A discussion after the surgical procedure among clinicians with different vision can result in valuable feedback for further development of the system, better acceptance of medical AR, detection of benefits for vertebroplasty and determination of further promising clinical applications. In addition, we believe that the concept of a *shadow surgeon* can have an important impact in teaching and training medical staff. This concept would allow us to perform clinical experimentation without the need for ethical approval and particular procedures for allowing such experiments.

3 FIRST FEASIBILITY STUDIES IN THE TRAUMA ROOM

In the first of two experimental setups, the *shadow surgeon* is positioned on the anesthesia site of the CT scanner, to watch the procedure through the gantry without restricting the freedom of movement of the performing surgeons (see Fig. 1(a)). Two sets of two tracking cameras are positioned on the opposite side of the CT scanner and locate the trokar through the gantry. A plastic spinal column, is tracked with non-sterile, non-certified, custom made, retro

* (bichlmei, aahmadi, navab)@cs.tum.edu

† (Ben.Ockert, Sandro-Michael.Heining)@med.uni-muenchen.de

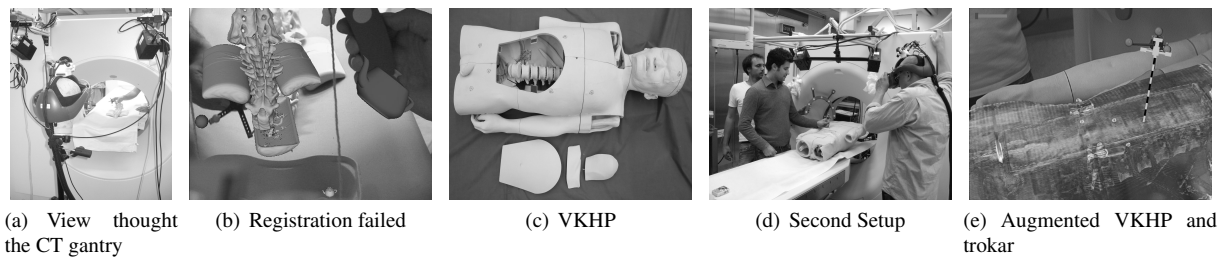


Figure 1: Figures (a),(b) show first setup, figures (d)(e) show second setup of feasibility studies in the trauma room. Figure (c) shows the VKHP.

reflective Beekley Corporation CT sphere markers attached to the bottom side of the bedding 1. This enables sterile and automatic registration of the imaging data with the patient's anatomy, however, a constant registration between bedding and patient can not be guaranteed. In the second experimental setup, the *shadow surgeon* is positioned on the same side of the CT scanner as the performing surgeons (see Fig. 1(d)). This enables a view on the patient corresponding to the perspectives of the performing surgeon. For this experiment, we attach the mentioned CT sphere markers to the skin surface of the VKHP. To evaluate the accessibility of imaging data from the CT scanner we superimpose the data from a scanned phantom instead of the original VKH CT data (see Fig. 1(e)). The VKHP and the trokar are tracked from a camera set installed above the position of the *shadow surgeon*. In both experiments, positioning the reference target used for single camera tracking turned out to be a major issue due to space restrictions within the field of view. For the upcoming experiments, we plan to attach a set of retro reflective flat disc markers to the CT scanner in order to replace the reference target.

4 DISCUSSION & FUTURE WORK

We can not expect that additional AR equipment is immediately accepted by the personnel in a highly organized, efficient environment that is scarce in space. We rather have to follow the strategy of finding niches within the OR to integrate smoothly our technology and letting the surgeons themselves detect and decide about the benefits of such systems and request for improvements. ARAV might not be the *killer application* to revolutionize the therapy. Nevertheless, it is perfectly suitable to analyze feasibility since it is a common standard procedure with a clearly specifiable workflow requiring frequent updates of imaging data. The operating environment is not highly staffed with standard equipment and surgical personnel, which leaves psychological and physical elbowroom to install the prototype AR system. Regarding accurate and sterile tracking, and registration for the minimal invasive vertebroplasty, we are currently investigating the integration of percutaneously tracking targets attached to the vertebrae. When tracking the bedding (section 3) we can not expect constant registration quality of imaging data, however, this concept provides a preliminary solution for sterile tracking of the patient. Unfortunately, the tracking cameras were not adjusted correctly to localize objects within the required distance, which results in bad registration quality (see Fig. 1(b)). *BrainLAB* and *Northern Digital Inc.* offer one-way passive sphere markers for high precision and sterile tracking of surgical tools, which were used in the second study to track the trokar. We also tested successfully autoclavable glass sphere markers developed at the chair for *Micro Technology and Medical Device Technology*, *TU München* with the outside in tracking system. Unfortunately, these markers are currently not available on the market. In our first experimental setup we found that the view of the *shadow surgeon* through the gantry of the CT scanner is too limited. In the subsequent brain storming, surgeons claimed that the *shadow surgeon* positioned on the same side of the CT scanner would not limit their

freedom of movement as long as the system and the *shadow surgeon* can be removed quickly from the field of action if necessary. This can be realized with a *hanging HMD* that can be lifted from the ceiling with a flexible arm construction. A primary solution for this is currently developed by our partners at Simiosys.

5 CONCLUSION

This paper gives an intermediate report on our project *ARAV - Augmented Reality Aided Vertebroplasty* with the objective of a permanent installation of an AR system in the OR. We presented different concepts, strategies and experiments in the trauma room for the transfer from lab space to the OR.

ACKNOWLEDGEMENTS

We would like to thank Frank Sauer from Siemens Corporate Research (SCR), Konrad Zürl and Oliver Wenisch from A.R.T. GmbH, Weilheim, the radiologists and surgeons of Klinikum Innenstadt München and Arno Scherhorn.

REFERENCES

- [1] S.-A. Ahmadi. Discovery and detection of surgical activity in percutaneous vertebroplasty. Master's thesis, Technische Universität München (TUM), 2008.
- [2] C. Bichlmeier, F. Wimmer, H. Sandro Michael, and N. Nassir. Contextual Anatomic Mimesis: Hybrid In-Situ Visualization Method for Improving Multi-Sensory Depth Perception in Medical Augmented Reality. In *Proceedings of the 6th International Symposium on Mixed and Augmented Reality (ISMAR)*, pages 129–138, Nov. 2007.
- [3] M. Figl, C. Ede, J. Hummel, F. Wanschitz, R. Ewers, H. Bergmann, and W. Birkfellner. A fully automated calibration method for an optical see-through head-mounted operating microscope with variable zoom and focus. *IEEE Trans. Med. Imag.*, 24(11):1492–1499, 2005.
- [4] W. E. L. Grimson, T. Lozano-Perez, W. M. Wells, III, G. J. Ettinger, S. J. White, and R. Kikinis. An automatic registration method for frameless stereotaxy, image guided surgery, and enhanced reality visualization. *IEEE Trans. Med. Imag.*, 15(2):129–140, 1996.
- [5] A. P. King, P. J. Edwards, C. R. Maurer, Jr., D. A. de Cunha, D. J. Hawkes, D. L. G. Hill, R. P. Gaston, M. R. Fenlon, A. J. Strong, C. L. Chandler, A. Richards, and M. J. Gleeson. Design and evaluation of a system for microscope-assisted guided interventions. *IEEE Trans. Med. Imag.*, 19(11):1082–1093, 2000.
- [6] N. Navab, J. Traub, T. Sielhorst, M. Feuerstein, and C. Bichlmeier. Action- and workflow-driven augmented reality for computer-aided medical procedures. *IEEE Computer Graphics and Applications*, 27(5):10–14, September/October 2007.
- [7] J. Park, M. Chung, S. Hwang, Y. Lee, D. Har, and H. Park. Visible korean human: Improved serially sectioned images of the entire body. 24(3):352–360, March 2005.
- [8] J. P. Rolland and H. Fuchs. Optical versus video see-through head-mounted displays in medical visualization. *Presence*, 9:287–309, 2000.
- [9] F. K. Wacker, S. Vogt, A. Khamene, J. A. Jesberger, S. G. Nour, D. R. Elgort, F. Sauer, J. L. Duerk, and J. S. Lewin. An augmented reality system for mr image - guided needle biopsy: Initial results in a swine model. *Radiology*, 238(2):497–504, 2006.